

CLOUD COMPUTING COURSE

Lecture1:Introduction to Cloud Computing

ABSTRACT
Lecture1 introduces cloud-computing definition, components and history.

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Introduction to Cloud Computing

-Cloud computing means that instead of all the computer hardware and software you're using sitting on your desktop, or somewhere inside your company's network, it's provided for you as a service by another company and accessed over the Internet, usually in a completely seamless way. Exactly where the hardware and software is located and how it all works doesn't matter to the user—it's just somewhere up in the "cloud" that the Internet represents.

-distributed computing on internet or delivery of computing service over the internet.

Example: Yahoo, Gmail, Hotmail: instead of running an e-mail program on your computer, you log in to a web e-mail account remotely. The software and storage for your account doesn't exist on your computer, it is on the services computer cloud.

Example: When you do a Google search, the real work in finding your answers might be done by a computer sitting in California, Dublin, Tokyo, or Beijing; you don't know—and most likely you don't care!

-Allows us to use IT services as conveniently as using utilities like water and electricity.

-Ready to use IT services over the internet.

Cloud Computing Examples

Examples: AWS, Azure, Google Cloud and HUWAEI CLOUD user portal-Products.

Example2: Cloud Computing Applications such as:

1. Backup and Restore application
2. Google Translate
3. iReader
4. Gmail: emails can be stored and processed through a server in some remote part of the world, easily accessible from a Web browser, wherever we happen to be. Pushing email off into the cloud makes it supremely convenient for busy people, constantly on the move.

5. SoundCloud: You can build up a personal collection of tracks you like and access them from any device, anytime, anywhere.

6. Google Documents: create a document, spreadsheet, presentation, or whatever you like using Web-based software. Instead of typing your words into a program like Microsoft Word or OpenOffice, running on your computer, you're using similar software running on a PC at one of Google's world-wide data centers.

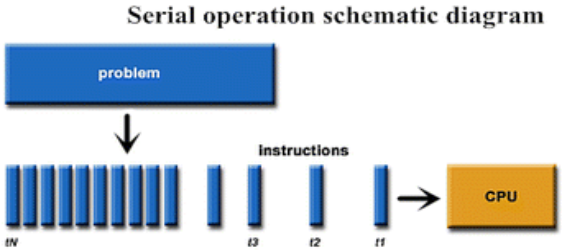
Computing History

1. Serial Computing

A processing in which one task is completed at a time and all the tasks are run by the processor in a sequence. This type of processing is also known as **sequential processing**.

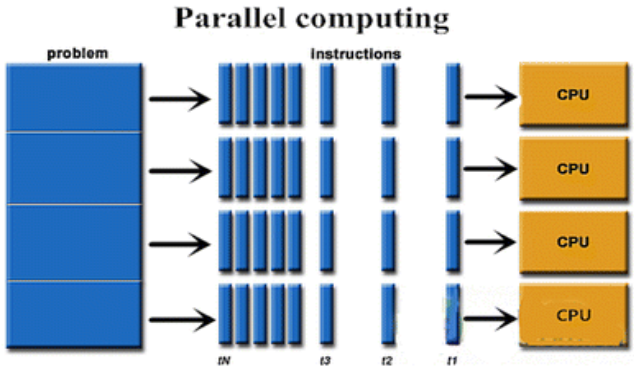
Example: Any operating system running on the single processor is an example of the serial operating system.

Pentium 3 and Pentium 4 are serial processing computers. Windows 95 and Windows 98 are examples of operating systems, which do the serial processing.



2. Parallel Computing

A type of processing in which multiple tasks are completed at a time by different processors.it consists of multiple processors that communicate with each other using a shared memory.



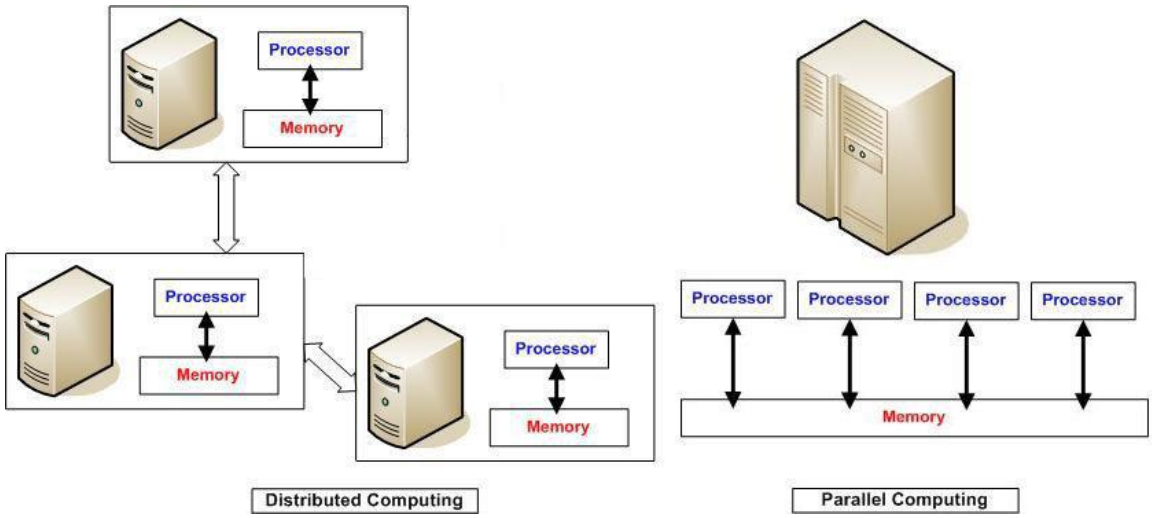
Example: An operating system running on the multicore processor is an example of the parallel operating system. Windows 7, 8, 10 are examples of operating systems which do parallel processing. In today life all latest operating systems support parallel processing.

3. Distributed Computing

It is a type of computing in which a group of computers are linked together so that they can act as a single entity. In distributed systems, there is no shared memory and computers communicate with each other through message passing.

Distributed computing divides a single task between multiple computers. Each computer can communicate with others via the network. All computers work together to achieve a common goal. Thus, they all work as a single entity. A computer in the distributed system is a node while a collection of nodes is a cluster.

It allows scalability and makes it easier to share resources easily. It also helps to perform computation tasks efficiently.



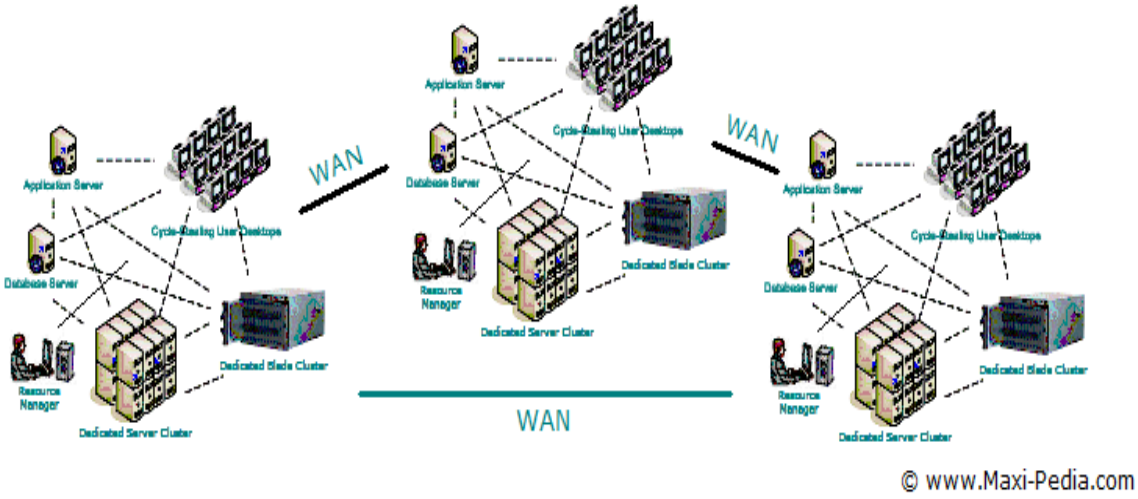
4. Grid Computing

It is a collection of computer resources from multiple locations to reach a common goal. grid works on various tasks within a network, but it is also capable of working on specialized applications. It is designed to solve problems that are too big for a supercomputer while maintaining the flexibility to process numerous smaller problems.

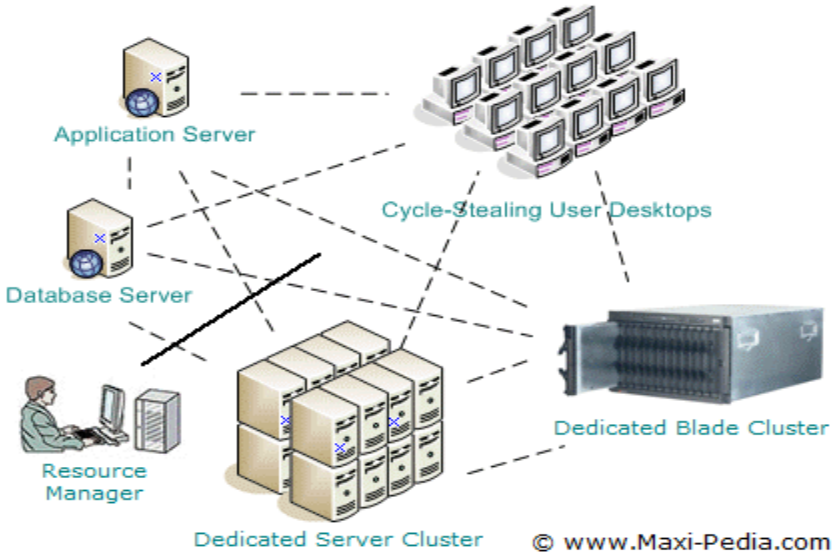
It is often used in structural analysis, Web services such as ATM banking, back-office infrastructures, and scientific or marketing research.

The key distinction between distributed computing and grid computing is mainly the way resources are managed: Distributed computing uses a centralized resource manager and

all nodes cooperatively work together as a single unified resource or a system while **Grid computing** utilizes a structure where each node has its own resource manager and the system does not act as a single unit.



Grid computing employs not only single resources but whole systems from various locations while crossing geographic and political boundaries.



The concept of distributed computing is simple -- pull together and employ all available resources to speed up computing.

5. Cloud Computing

It is a type of computing that relies on sharing computer resources rather than having a local servers or personal devices to handle applications.

Cloud Computing Definition

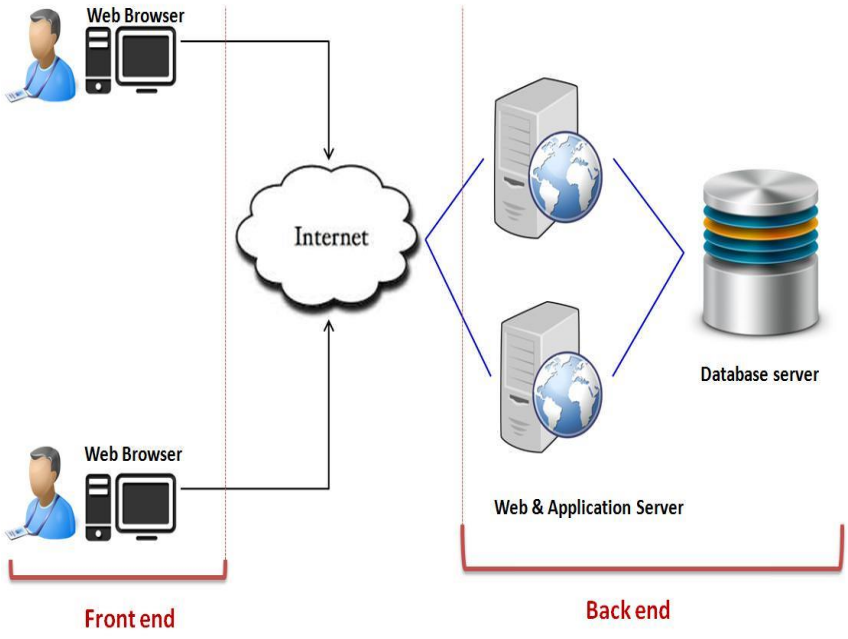
Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

Cloud Computing Architecture

Cloud computing comprises of two components:

1. **1. Front End**: it consist client part of cloud computing system (PC, Laptop or Mobile). It comprise of **interfaces and applications** that are required to access the cloud-computing platform.
2. **2. Back End:** refers to the cloud itself, it comprises of the resources that are required for cloud computing services. It consists of **virtual machines, servers, data storage, security mechanism etc.** It is under provider's control.

Both of a component of cloud computing are connected via a network, usually the Internet. In a cloud computing architecture, all applications are controlled, managed, and served by a cloud server. Its data is replicated and preserved remotely as part of the cloud configuration over multiple hard disks and machines.



There are certain services and deployment models working behind the scenes that make cloud computing feasible and accessible to the end users.